Example 2.25 Resolve into partial fractions: $\frac{x}{(x+3)(x-4)}$.

Solution:

Let $\frac{x}{(x+3)(x-4)} = \frac{A}{x+3} + \frac{B}{x-4}$ where A and B are constants.

Then, $\frac{x}{(x+3)(x-4)} = \frac{A(x-4)+B(x+3)}{(x+3)(x-4)}$, which gives $x = A(x-4)+B(x+3)$.

When $x = 4$, we have $B = \frac{4}{7}$.

When $x = -3$, we have $A = \frac{3}{7}$

Hence, $\frac{x}{(x+3)(x-4)} = \frac{3}{7(x+3)} + \frac{4}{7(x-4)}$.



The above procedure can be carried out if the denominator has all its zeros in $\mathbb{R}$ which are all distinct.

Example 2.26 Resolve into partial fractions: $\frac{2x}{(x^2+1)(x-1)}$.

Solution:

In this case, note that the denominator has a factor x^2+1 which does not have real zeros.

Let $\frac{2x}{(x^2+1)(x-1)} = \frac{A}{(x-1)} + \frac{Bx+C}{x^2+1}$
where A, B, C are constants.

We have, $2x = A(x^2+1) + (Bx+C)(x-1)$.

When $x = 1$, we get $A = 1$.

When $x = 0$, we have $A - C = 0$ and hence $A = C = 1$.

When $x = -1$, we have $2A - 2(C-B) = -2$, which gives $B = -1$.

Thus, $\frac{2x}{(x^2+1)(x-1)} = \frac{1}{(x-1)} + \frac{1-x}{x^2+1}$

We now illustrate the situation when denominator has a real zeros with multiplicity more than one.

Example 2.27 Resolve into partial fractions: $\frac{x+1}{x^2(x-1)}$.

Solution:

Let $\frac{x+1}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1}$.

Then, $x+1 = Ax(x-1) + B(x-1) + Cx^2$.

When $x = 0$, we have $B = -1$ and when $x = 1$, we get $C = 2$.

When $x = -1$, we have $2A - 2B + C = 0$ which gives $A = -2$.

Thus, $\frac{x+1}{x^2(x-1)} = \frac{-2}{x} - \frac{1}{x^2} + \frac{2}{x-1}$.